

1 Easy

Problem 1 (Source: ???)

Jessica fills the 64 squares of an 8×8 table with distinct positive integers from 1 to 66 in a way such that the sum of each row is divisible by 17 and the sum of each column is divisible by 21. What is the sum of the two positive integers she omits?

2 Easy

Problem 2 (Source: ???)

Compute the number of 6-digit positive integers $\overline{ABCDEF}$ satisfying:

- $\overline{AB}$ is divisible by 4
- $\overline{ABCD}$ is divisible by 16
- $\overline{ABCDEF}$ is divisible by 64

3 Medium

Problem 3 (Source: ???)

Call an ordered pair of positive integers (a, b) *spooky* if it satisfies

$$\gcd(a, b)^2 = \text{lcm}(a, b) + 2023.$$

What is the number of spooky pairs of positive integers?

4 Medium

Problem 4 (Source: ???)

Find all real numbers m such that $m^2 - \frac{1}{m^2}$ and $m^3 - \frac{1}{m^3}$ are integers.

5 Hard

Problem 5 (Source: ???)

Sequences a_n, b_n with $a_1 = b_1 = 1$ satisfy the recurrences $a_{n+1} = a_n + 2b_n$ and $b_{n+1} = a_n + b_n$. Find the remainder when $a_{306} \cdot b_{306}$ is divided by 17.

6 Hard

Problem 6 (Source: ???)

Consider function f satisfying $f(0) = 0$ and $f(n) = f(n - p) + 1$ for all integers $n > 1$, where p is the smallest prime divisor of n . Find the sum of all k such that the number of composite solutions to $f(n) = k$, $1 < n \leq 500$ is maximized.

7 Hard

Problem 7 (Source: ???)

Find the number of distinct values that

$$\frac{i^4 - 1}{i^2}$$

takes modulo 2023 over all i such that $\gcd(i, 2023) = 1$.